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Ethical Dilemmas Facing Nephrology

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CHAPTER OUTLINE

KEY POINTS

- Ethical issues facing nephrology played a fundamental role in early development of the field of bioethics. Because dialysis and kidney transplantation predated many other types of life-sustaining therapies, many of the ethical issues that inevitably accompany highly intensive medical treatments were first encountered by the kidney community.
- In the 1960s, the Seattle Artificial Kidney Center Admissions and Policy Committee used "social worth" criteria to select patients to receive scarce and life-saving hemodialysis treatments. This practice was widely criticized and directed public attention to bioethical issues in medicine.
- The U.S. Medicare End-Stage Renal Disease program was developed in 1972 to support treatments for kidney failure for U.S. citizens, but costs quickly expanded beyond initial projections. Shifting costs reflected, in part, changing demographics of the dialysis population to include older adults and people with complex comorbidities.
- Advances in postnatal care have greatly improved outcomes for very premature infants with renal dysfunction and severe cognitive deficits. These technologies raise questions about how to synthesize goals for longevity and quality of life for these young patients.
- Shared decision making is the recommended approach to guide decisions about kidney failure treatments—including dialysis, active medical management without dialysis (also known as *comprehensive conservative care* or *conservative kidney management*), and transplant—that align with an individual patient's preferences, goals, and values.
- Advance care planning and the documentation of advance directives is especially important for patients receiving dialysis, many of whom are older adults with limited prognosis. Prior discussion of advance directives has been shown to help patients and their families approach end of life in a way that upholds their own values.
- Initiating dialysis as a "time-limited trial" can be a way to avoid a sense of inflexible commitment to a treatment strategy, especially for seriously ill patients for whom the benefit of treatment may be uncertain. Dialysis may be initiated with a planned time to reevaluate whether goals are being met.

KEY POINTS—cont'd

- Major disparities in access to kidney transplantation continues. Black patients are less likely than White patients to be referred for transplant evaluation, to complete a transplant evaluation, to be added to a transplant waiting list, and to receive a kidney transplant. Revisions of the national rules for deceased donor kidney allocation in 2014 were intended to improve both equity and effectiveness of the process. Ongoing work is needed to address barriers to the transplant evaluation and selection process that occur long before waitlisting.
- The cost of care for Americans with kidney failure has far outstripped projections from when the end-stage renal disease Medicare entitlement was enacted. Medicare has trialed multiple value-based payment models intended to support quality of care for this population while containing costs. Payment models have been designed to support care along the spectrum of kidney disease including prevention of progression to kidney failure, to support integration of care between providers, and to encourage adoption of home dialysis modalities and transplantation.
- Innovative new technologies and care processes are needed to improve the health and experience of patients with kidney disease, many of whom continue to endure burdensome dialysis treatments. However, technologic advances should be accompanied by intentional and proactive bioethical analysis and community engagement to determine fair and respectful application of these technologies.

LEGACY OF BIOETHICAL ISSUES IN KIDNEY CARE

NEPHROLOGY'S CONTRIBUTION TO THE BIRTH OF BIOETHICS

Any review of ethical issues facing nephrology must begin with the fundamental role that nephrology played in the early development of the field of bioethics. Because dialysis and kidney transplantation predated many other organ-replacement therapies, many of the ethical issues that inevitably accompanied these highly specialized medical treatments were first addressed by nephrologists. This chapter focuses on ethical issues in the treatment of kidney failure viewed through the lens of experience in the United States—the location of many early innovations—but a global perspective reveals the additional complexity of treating patients with advanced kidney disease in differing health care systems and underresourced settings. This broader perspective is addressed briefly at the conclusion of the chapter.

In 1960, maintenance hemodialysis (also called *chronic* or *long-term hemodialysis*) first became feasible with the invention of the arteriovenous shunt. Amid the excitement of this medical advance, nephrologists were quickly confronted by the unprecedented ethical problem of determining which patients should be granted access to this life-saving therapy that was limited by the availability of dialysis machines. Resource allocation decisions extended beyond clinical facts to include value judgments and issues of justice. One early U.S. dialysis center, the Seattle Artificial Kidney Center, invited a group of local citizens to make allocation decisions, a role that was previously well within the guarded realm of physician autonomy. Historian David Rothman and bioethics scholar Albert Jonsen heralded this engagement of the lay population as the event that signaled the entrance of bioethics into medicine.[1,2]

In his 1964 presidential address to the American Society of Artificial Internal Organs, Dr. Belding Scribner, one of the early fathers of nephrology, identified four major ethical problems that he and his colleagues were facing: 1. Patient selection for dialysis; 2. Termination of dialysis, which he called "dialysis suicide" but is now referred to as "dialysis discontinuation," "stopping dialysis," or "withdrawing dialysis"; 3. "Death with dignity," which involved the treatment of patients with kidney failure when dialysis was withheld or withdrawn; and 4. Donor selection for transplantation.[3] Indeed, Dr. Scribner anticipated many of the ethical challenges that would continue to face nephrology—as well as other fields of medicine—for decades.

Ethical issues arising in treating patients with advanced chronic kidney disease (CKD) can be unpacked by applying four ethical principles: respect for autonomy, beneficence, nonmaleficence, and justice.[4] This widely accepted framework, called *principalism,* is used to identify and analyze ethical problems in medicine. Fig. 80.1 presents a timeline for the relative salience of these principles throughout the history of the end-stage renal disease (ESRD) program.[4]

EARLY ACCESS TO SCARCE MAINTENANCE HEMODIALYSIS TREATMENTS AND PATIENT SELECTION CRITERIA

The invention of the Quinton–Scribner arteriovenous shunt offered the ability to repeatedly access a patient's blood circulation over regular dialysis treatments, making maintenance dialysis for chronic kidney failure (vs. treatment only for acute kidney failure) feasible for the first time. When the first U.S. maintenance dialysis center, the Seattle Artificial Kidney Center in the U.S. Pacific Northwest, opened in 1962, the number of patients seeking dialysis treatment far outweighed the facility's ability to provide it. Physicians used clinical criteria to narrow the group of

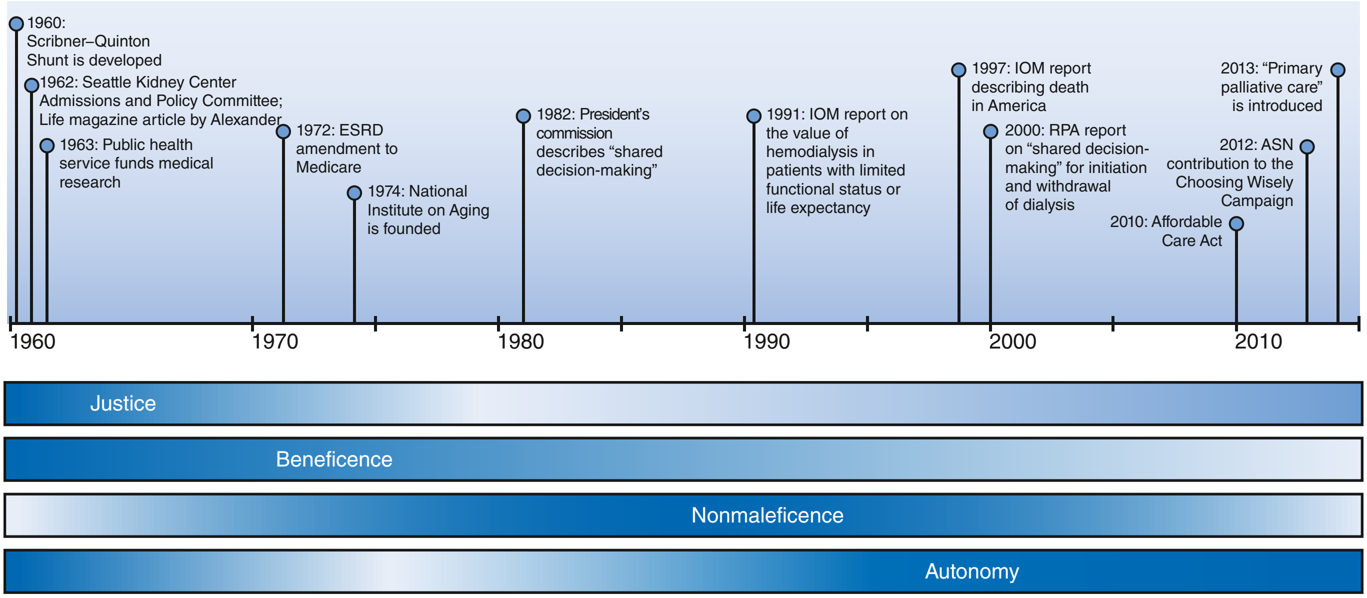

Fig. 80.1 Timeline of important events and publications related to the development of dialysis. Although there is significant overlap, the relative salience of each of the four principles varies over this course. *ASN,* American Society of Nephrology; *ESRD,* end-stage renal disease; *IOM,* Institute of Medicine; *RPA,* Renal Physicians Association. (Used with permission of the American Society of Nephrology. Prior publication in the *Clinical Journal of the American Society of Nephrology*: Butler CR, Mehotra R, Tonelli MR, Lam DY. The evolving ethics of dialysis in the United States: a principlist bioethics approach. *Clin J Am Soc Nephrol*. 2016;11:704–709.)

potential patients to otherwise healthy adults aged 18 to 45 years, but there remained more candidates than could be accommodated and no established criteria to decide who among them would receive the life-sustaining treatment.

Health care rationing occurs when a treatment that may benefit a patient is withheld because of resource limitations. The ethical principle of justice requires distribution of societal resources in a fair, transparent, and consistent way. In an attempt to integrate public values about fair allocation into their decision-making process, the Seattle group appointed the Admissions and Policy Committee, a group of citizens who were tasked with representing the local community in selecting medically suitable dialysis candidates. The committee initially considered choosing patients by lottery but rejected this idea, believing that more ethical decisions could be made by active selection on the basis of a patient's ability to be rehabilitated and eventually return on society's investment in a patient's health. Factors taken into consideration included age and gender, marital status and number of dependents, income, net worth, emotional stability, educational background, occupation, and future potential contribution. As the selection process evolved, a pattern emerged in which committee decisions weighed heavily on a narrow definition of a favorable personal character and valuable contribution to society.[5]

Journalist Shana Alexander called national attention to the committee in her *Life Magazine* article, "They Decide Who Lives, Who Dies: Medical Miracle Puts Moral Burden on Small Committee."[5] This publicity precipitated a broader public outcry. The committee was castigated for using upper middle-class values and social-worth criteria to make life-and-death decisions.[6] Critics wrote, " The Pacific Northwest is no place for a Henry David Thoreau with bad kidneys."[7] The complex ethical structure guiding medical decision making was exposed to the public, making several points clear. First, it is unacceptable to make resource allocation or rationing decisions on the basis of social worth. More broadly, it became clear that the public has a moral stake in how clinical decisions are made.

U.S. HEALTH CARE COVERAGE FOR DIALYSIS AND AN EXPANDING PATIENT POPULATION

In 1972, passage of the ESRD Amendment to Medicare (Section 299I of Public Law 92-603) aimed to eliminate the need to ration dialysis. The U.S. Congress passed this legislation compelled by the life-and-death need for dialysis with an expectation that the number of patients who would need this treatment would be limited. This legislation extended coverage of treatments for kidney failure to Americans who were otherwise eligible for Social Security.[8] Neither the medical community nor Congress believed it to be necessary or proper for the government to further direct decisions about which patients were appropriate candidates for dialysis. Indeed, while early access to dialysis had been limited to young and relatively healthy patients, many believed that it was morally unjustified to deny dialysis treatment to any patient with kidney failure.[6] A decade later, the first report of the U.S. Renal Data System documented an escalating incidence of patients initiating dialysis,[9] explained in part by the growing population of older adult patients and those with comorbidities who were receiving the treatment.[10] By 2000, 48% of incident patients receiving dialysis were 65 years of age or older and 45% had kidney failure from diabetes. Observers raised concerns about the appropriateness of starting patients on dialysis with a limited life expectancy and uncertain quality of life.[8,11] Others questioned the utility of providing dialysis to patients with life-threatening illnesses such as end-stage cardiovascular disease, debilitating pulmonary disease, or untreatable cancer. Benefit of the treatment was also unclear in patients with severe neurologic disease, which rendered them unable to engage with others, such as those in a persistent vegetative state, with severe dementia, or with extensive cerebrovascular disease.[12]

In 1991 the Institute of Medicine (now National Academy of Medicine) Committee for the Study of the Medicare ESRD Program emphasized that the existence of the public entitlement did not obligate physicians to treat all patients with kidney failure with dialysis or transplantation.[8] For some, the burden of dialysis substantially outweighed the benefit. At the same time, systematic differences between providers in determining the relative benefit and burden of the treatment for patients risked exacerbating disparity in access to health care and health outcomes.[13,14] The committee recommended that guidelines be developed to help nephrologists make dialysis decisions more uniformly, with greater ease, and in a way that promoted patient benefit. In response, the American Society of Nephrology (ASN) and the Renal Physicians Association published the clinical practice guideline, *Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis,* in 2000, which was updated in 2010.[15]

ETHICAL ISSUES IN DIALYSIS TREATMENT FOR KIDNEY FAILURE

WITHHOLDING AND WITHDRAWING DIALYSIS

In the 1960s, patients accepted for dialysis felt fortunate to receive a scarce therapy. When some patients started to choose to forgo this life-saving treatment, pleas to "stop the machine" were initially termed "dialysis suicide" and framed as a form of psychopathology. However, drawing on Catholic moral theology, clinicians and ethicists began to reconceptualize decisions to refuse or stop aggressive interventions such as dialysis, which represent nonobligatory extraordinary (versus ordinary) attempts to sustain life.[16] In its 1983 report, *Deciding to Forego Life-Sustaining Treatment,* the President's Commission for the Study of Ethical Problems in Medicine and Biomedical and Behavioral Research considered ethical and legal issues raised by life-sustaining treatments such as dialysis and reached two major conclusions that provide justification for withholding and withdrawing dialysis: "The voluntary choice of a competent and informed patient should determine whether or not life-sustaining therapy will be undertaken;" and "Health-care professionals serve patients best by maintaining a presumption in favor of sustaining life, while recognizing that competent patients are entitled to choose to forgo any treatments, including those that sustain life."[17] Active withdrawal from dialysis now represents approximately 17% of deaths among people with kidney failure receiving hemodialysis in the United States.[18]

Shifting regulations and clinical practice guidelines emphasized the importance of optimizing benefits and reducing harm while also respecting patients' autonomous choice. However, the best process to develop a balance between these values and incorporate perspectives from the patient, family, and nephrologist remained opaque. Individual patients evaluate benefits and burdens differently. Some patients have limited cognitive capacity and require surrogates to support decision making. Box 80.1 provides recommendations for a systematic approach for responding to a patient request to stop dialysis. It recognizes the right of cognitively competent patients or the surrogate of patients without cognitive capacity to make this decision for themselves but, at the same time, urges a careful review for

Box 80.1 Systematic Evaluation of a Patient or Family Request to Stop Dialysis

1. Determine the reasons or conditions underlying the patient/surrogate desires regarding withdrawal from dialysis. Such assessment should include specific medical, physical, spiritual, and psychological issues, as well as interventions that could be appropriate. Some of the potentially treatable factors that might be included in the assessment are as follows:
   - Underlying medical disorders including a prognosis for short- or long-term survival on dialysis
   - Difficulties with dialysis treatments
   - The patient's assessment of his or her quality of life and ability to function
   - The patient's short- and long-term goals
   - The burden that costs of continued treatment/medications/diet/transportation may have on the patient/family/others
   - The patient's psychological condition including depression and conditions or symptoms that may be caused by uremia
   - Undue influence or pressure from outside sources including the patient's family
   - Conflict between the patient and others
   - Dissatisfaction with the dialysis modality, time, or setting of treatment
2. If the patient wishes to withdraw from dialysis, did he or she consent to referral to a counseling professional (e.g., social worker, spiritual advisor, psychologist, psychiatrist)?
3. If the patient wishes to withdraw from dialysis, are there interventions that could alter the patient's circumstances that might result in his or her considering it reasonable to continue dialysis?
   - Describe possible interventions.
   - Does the patient desire the proposed intervention(s)?
4. In cases in which the surrogate has made the decision to either continue or withdraw dialysis, has it been determined that the judgment of the surrogate is consistent with the stated desires of the patient?
5. Questions to consider when a patient asks to stop dialysis:
   - Is the patient's decision-making capacity diminished by depression, encephalopathy, or other disorder?
   - Why does the patient want to stop dialysis?
   - Are the patient's perceptions about the technical or quality-of-life aspects of dialysis accurate?
   - Does the patient really mean what he or she says, or is the decision to stop dialysis made to get attention, help, or control?
   - Can any changes be made that might improve life on dialysis for the patient?
   - Would the patient be willing to continue dialysis while the factors responsible for the patient's request are addressed?
   - Has the patient discussed his or her desire to stop dialysis with significant others such as family, friends, or spiritual advisors? What do they think about the patient's request?

From Shared Decision-Making in the Appropriate Initiation of and Withdrawal from Dialysis. 2nd ed. Rockville, MD: Renal Physicians Association; 2010.

any mediating medical, physical, spiritual, and psychological factors that might be addressed to improve patients' quality of life before deciding to stop dialysis.

KIDNEY CARE FOR A GROWING POPULATION OF OLDER ADULTS

Older age and medical comorbidity are now rarely perceived to be contraindications to dialysis. Far from it, as the U.S. population ages and a range of chronic conditions become more prevalent, those aged 75 years or older have become the fastest growing component of the dialysis population.[18] Between 1996 and 2003, rates of dialysis initiation among octogenarians and nonagenarians nearly doubled.[19] Supporting older adults in deciding whether to start dialysis involves considering factors including reduced life span, complex comorbidities, heterogeneous personal values, and varied quality of life.[20]

Characteristics such as older age, nonambulatory status, poor nutritional status, and presence of comorbid conditions are associated with shorter life spans.[19] Among U.S. veterans with advanced CKD, adults older than 75 years were far more likely to die than develop kidney failure in their lifetime.[21] In multiple studies comparing dialysis with active medical management without dialysis for patients older than 75 years of age with advanced CKD, dialysis has not been associated with a survival benefit for patients with ischemic heart disease or those with the highest burden of comorbidities.[22,23] Older adults with advanced kidney disease and dementia may be especially unlikely to benefit from dialysis for the following reasons: 1. They may not live any longer with dialysis than without it, especially if they have other comorbidities; 2. They have a twofold higher mortality risk than those without dementia; 3. They are less likely to do well on dialysis and more likely to stop it; 4. The dialysis center is a "dementia-unfriendly" environment, and these patients may not be able to cooperate safely with the dialysis process; and 5. They experience a significant decline in cognitive function, particularly executive function, after hemodialysis initiation.[24]

Given the unique aspects of caring for an older and multimorbid population, the Renal Physicians Association and American Society of Nephrology clinical practice guideline emphasized the importance of discussing prognosis and conducting advance care planning with older adult patients and their families to make individualized decisions.[15,19] These recommendations argue against an "age-neutral" approach to the management of CKD and emphasized the need for prognostic tools that enable clinicians to identify the subgroup of older adults who are most likely to benefit from dialysis.[21] A potentially helpful prognostic tool to estimate 6-month mortality in dialysis patients considers age, serum albumin, dementia, peripheral vascular disease, and the nephrologist's response to the surprise question ("Would I be surprised if this patient died in the next 6 months?")[25] Another prognostic tool was developed using the French Renal Epidemiology and Information Network and estimates 90-day mortality in patients 75 years or older who are starting dialysis and integrates risk factors including age over 85 years, cardiovascular comorbidities, cognitive disorders, active malignancy, serum albumin, and impaired mobility.[26] Some have questioned the standard content of informed consent for dialysis and advocate for an age-sensitive approach that is attuned to the different balance of benefits and harms of the treatment for older adults with multiple comorbidities.[27] The American Geriatrics Society has developed a structure for approaching the medical evidence base, recognizing that it is often difficult to extrapolate results from trial data to an older population.[28] See Chapter 60 for more discussion of decision making and clinical management of older patients with CKD.

Clinical Relevance

Characteristics of patients with advanced CKD such as older age, poor functional status, poor nutritional status, dementia, frailty, and high burden of comorbidity score are associated with shorter life span and poorer outcomes. Dialysis may not offer greater survival compared with active medical management without dialysis for many patients. Clinicians should include prognostic considerations in discussions about treatment options.

KIDNEY CARE FOR VERY PREMATURE INFANTS

Young children are assumed to lack capacity to make autonomous decisions, and medical decisions are typically deferred to parents who are expected to act in the child's best interests. These best interests often relate to preserving the child's ability to become an autonomous adult and ultimately make his or her own life decisions. This protective stance in the context of a Western cultural default to life-extending interventions, as well as the tragedy of childhood illness and premature death, supports a strong tendency toward intensive medical interventions for children with kidney disease. However, this strong drive to protect pediatric patients above other considerations may detract from other important goals such as time with family, play, and life participation.[29] This focus also likely contributes to a limited participation of children with kidney disease in clinical trials and subsequent dearth of an evidence base.[30]

Advances in life-sustaining technology, nutrition, and nursing practices allow for survival of increasingly premature infants, but prematurity nonetheless is associated with a range of acute and chronic clinical conditions and survival can necessitate a range of complex and intensive interventions early in life. Premature infants may be born with congenital renal abnormalities and kidney failure and are also at higher risk of intracerebral hemorrhage and cognitive deficits. In addition to a surgical procedure to place dialysis access and regular dialysis treatments, a premature infant with kidney failure is likely to require artificial nutrition, frequent blood draws, a range of medications, and hospitalizations for complications of their health conditions.[31] Indeed, while dialysis may sustain life, many nephrologists struggle with the ethical implications of potential suffering and recognize the need for individualized ethical deliberation that incorporates not only clinical considerations but family and cultural values grounded in the details of each individual case.[32]

Maintenance dialysis is also unique compared with other intensive in-patient interventions in that it is ultimately a home-based and long-term therapy that can require sustained and extraordinary efforts of parents and other caregivers.[31] Many parents of young children with kidney failure find value and meaning in caregiving, but the experience can also be physically and emotionally intensive and draw parents' resources away from other children, family members, and life goals.[33] Traditional clinical bioethics strongly emphasize a clinician's primary responsibility to the patient, but the substantial impact that dialysis treatments have on a dependent child's family calls into question this paradigm. The European Paediatric Dialysis Working Group summarized the complex factors to consider in initiating or withholding dialysis for pediatric patients including "comorbidities and predicted future quality of life for the child and family; availability of resources, both medical and those of the family support structure; and prognosis and possibility for future transplantation."[31]

BARRIERS TO ACTIVE MEDICAL MANAGEMENT WITHOUT DIALYSIS

For the many patients for whom dialysis offers equivocal benefits and/or does not support their life goals, active medical management without dialysis is a viable treatment option. In a study investigating the treatment of active medical management without dialysis, older patients with advanced CKD survived a median of 15 months from the point that their eGFR reached <15 mL/min/1.73 m^2. When supported by a multidisciplinary palliative care program, these patients experienced improvement in symptom burden and half the rate of hospitalization compared with patients who started dialysis and who were on average 10 years younger.[34] However, despite a growing evidence base suggesting that dialysis may not benefit all patients and many would prefer active medical management without dialysis, other individual and system-level forces may shape decisions to initiate dialysis.[35]

Nephrologists, who may view their role as providing active intervention and instilling hope, may perceive medical management without dialysis to be equivalent to "no care."[36] These providers may implicitly or explicitly take on a paternalistic stance in which they act to support what they perceive to be a patient's best interest. Persistent educational gaps among nephrologists about how to provide care without dialysis also act as barriers to alternative treatments. They may also be swayed by the norms and culture of their institution propelling patients toward in-center hemodialysis.[37]

A qualitative study of documentation in the U.S. Veterans Affairs medical record for veterans with advanced kidney disease who had chosen *not* to start dialysis suggested that patients would be repeatedly questioned about this decision with the expectation that they were likely to change their minds.[38] Initiation of dialysis served as a powerful default option with few perceived alternatives. Patients might experience "immense pressure" to initiate dialysis. These findings resonate with work of the medical anthropologist Sharon Kaufman, who describes the "routinization" of starting dialysis and how these practices are shaped by clinicians and societal expectations about intensive interventions as "ordinary" treatments.[39] She reported that nephrologists may feel obligated to dialyze adult patients even if they have personal moral concerns about whether the patient is likely to benefit.[40] The power of defaults illuminates care processes in medicine more broadly and underlines the importance of engaging patients proactively, consciously, and deliberately in care decisions.[41]

A powerful default to initiate dialysis limits patients' opportunity to make informed choices. To counter this default and support shared decision making, the kidney community needs to make progress in the following areas: 1. Raise awareness among nephrologists and develop a community buy-in that dialysis is not the best option for all patients and that it is the patient's right to decide among multiple treatment options, 2. Develop a clear pathway for treating patients who opt for active medical management without dialysis that is supported by regulatory policy and sufficient compensation to clinicians, and 3. Train nephrology clinicians in communication skills so that they may confidently and comfortably engage patients in the process of shared decision making about the range of treatment options.

SHARED DECISION MAKING ABOUT TREATMENTS FOR KIDNEY FAILURE

Shared decision making is a process in which clinicians elicit individual patient preferences, needs, and values to determine an approach to treatment that best respects these values and preferences,[42] and it is a valuable strategy to support person-centered care. In 2012 as one of its five recommendations in the American Board of Internal Medicine's Choosing Wisely Campaign, the ASN urged nephrologists and patients: "Don't initiate chronic dialysis without ensuring a shared decision-making process between patients, their families, and their physicians."[43] The ASN embraced shared decision making as the "preferred model for medical decision-making because it addresses the ethical need to fully inform patients about the risks and benefits of treatments, as well as the need to ensure that patients' values and preferences play a prominent role."[15]

Despite this recommendation, shared decision making continues to be poorly implemented into decision making about treatments for kidney failure.[44] The SDM-Q-9 tool was developed to measure the presence and quality of shared decision making in real-world practice. The tool measures 1. whether a patient is aware of choices and the decision to be made; 2. if the patient has an opportunity to express needs, values, and preferences; 3. whether the patient is informed of associated prognosis and benefits and risks of each option; 4. use of decision aids to standardize information and facilitate understanding; 5. adequacy of time for decision making; 6. whether the patient is an active participant, and 7. whether the patient is a coequal with the physician in the treatment decision and in developing a plan of care.[45] Participants with CKD in a study using the SDM-Q-9 commonly rated the shared decision making process as "suboptimal."[46]

Shared decision making can be strengthened using patient decision aids and other communication tools. Patient decision aids are intended to help patients prepare for a collaborative role in shared decision making with their providers, understand their options, clarify their personal values and apply these to potential outcomes for each option, and communicate their values and preferences to their providers.[47] There is increasing evidence for the value of decision aids to assist patients in evaluating options for kidney failure treatment including hemodialysis and peritoneal dialysis (PD) at home or in-center, kidney transplantation, a time-limited trial of dialysis, palliative dialysis, active medical management without dialysis, or deciding not to decide about dialysis.[48]

> ***Clinical Relevance***
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> A process of shared decision making has become the ethical standard for clinical decision making that supports person-centeredness. Through this process, patients and their care providers agree on an approach to treatment that is grounded in a common understanding of the patient's values and goals, the patient's overall condition and prognosis, and the risks and benefits of each treatment option. The American Society of Nephrology and the Renal Physicians Association have emphasized the critical importance of shared decision making before the initiation of dialysis.

PALLIATIVE AND END-OF-LIFE CARE FOR PATIENTS WITH KIDNEY DISEASE

In light of the limited prognosis and high symptom burden for people with kidney failure, multiple leaders and institutions have emphasized the importance of integrating palliative care and advanced care planning for end-of-life care into routine kidney care processes (for further discussion on supportive care, see Chapter 61).[49,50] Nephrology trainees also report a sense of moral obligation to provide such care,[51] though nephrology training programs still include limited education in primary palliative care and end-of-life care.[52,53]

By the 1990s, surveys suggested that 90% of nephrologists would stop dialysis at the request of a patient with decision-making capacity.[13,14] However, they disagreed about how to treat these patients in the absence of clear advance directives.[14] A patient's right to forgo dialysis was difficult to exercise in these circumstances and often overruled by a strong default to continue dialysis. For these reasons, nephrologists have been encouraged to discuss the circumstances under which patients would want to discontinue dialysis and forgo cardiopulmonary resuscitation and to urge patients to complete written advance directives.[54] These practices are intended to better insure that approach to care aligns with patient preference and supports patient autonomy in the event that a patient is no longer cognitively able to participate in care discussion.

The Patient Self-Determination Act of 1991 required institutions participating in Medicare and Medicaid to educate patients about the importance of completing advance directives.[55] Although the ESRD program was almost entirely funded by Medicare, dialysis units were inadvertently left out of the legislation. This was a problematic omission given that advance care planning is especially important for dialysis patients for several reasons[56]: 1. Older adults make up about half of the dialysis population, and these patients have a relatively short life expectancy on dialysis and are the most likely to withdraw or be withdrawn from the treatment; 2. Prior discussion of advance directives has been shown to help patients with kidney failure and their families approach death in a reconciled fashion[57]; 3. Patients who complete advance directives are more likely to have their wish to die at home respected; 4. Cardiopulmonary resuscitation rarely extends survival for seriously ill patients receiving dialysis but is typically provided by default in the absence of a specific directive to withhold this intensive intervention.[58]

Prognostication is a centerpiece of advance care planning discussions but notoriously difficult for clinicians. The "surprise" question identifies dialysis patients at high risk for early mortality. The odds of a patient dying within a year were shown to be 3.5 times higher if the nephrologist or nephrology nurse practitioner answered "No" than for those patients about whom the provider answered "Yes."[59] Targeting patients identified as seriously ill using the surprise question for goals of care discussions led to a 35% increase in complete advance care planning documentation in one study.[60]

Even with effective communication and preparation, it is often difficult for patients to fully anticipate the impact of dialysis on their life. Planning for a "time-limited trial" of dialysis may reduce a sense of inflexible commitment to a treatment strategy when the benefit of the treatment to the particular patient is uncertain.[61] Dialysis may be initiated with a planned reevaluation of whether goals are being met with continuation or withdrawal based on this future assessment. Patients may also reasonably prefer to decide not to decide.[62] The "deciding not to decide" option gives older adults with kidney disease more time and space to consider bigger picture goals and preferences about how to live their lives by deferring a decision about treatment options for kidney failure with their nephrologists until a mutually agreeable time.

ETHICAL ISSUES IN KIDNEY TRANSPLANTATION

ACCESS TO RENAL TRANSPLANTATION IN THE UNITED STATES

Dr. Scribner was prescient in 1964 when he identified patient selection for kidney transplantation as one of the major ethical problems facing nephrology. Indeed, the kidney transplant community continues to face pervasive ethical dilemmas around how to both maximize the benefit of a scarce supply of donor organs and at the same time support equitable distribution of this public resource across a population. Further, the community must determine whether and when to allow a healthy living donor to undergo a major surgical procedure without expectation of direct clinical benefit to themselves and how to respect living and deceased donors who offer the unique "gift" of a transplantable organ.

Kidney transplants can offer substantial benefits in terms of quality of life and increased longevity compared with dialysis, and many patients prefer this treatment approach.[63] While for some patients, the risks of transplant outweigh benefits, advances in surgical technology and immunosuppressive management mean that transplant can benefit an increasingly broad swath of people with kidney failure including frail older adults and people with multiple comorbidities.[64,65] Transplantation also confers lower costs to the health care system compared with a lifetime of dialysis treatments. However, a shortage of deceased donor organs means that the need for transplant far outstrips available resources. In 2020, more than 130,000 people progressed to kidney failure in the United States, but there were only about 25,000 kidney transplant procedures.[18] In this year more than 75,000 people who had been deemed appropriate candidates for transplant were on the national deceased donor kidney waiting list. After 3 years, a third of patients were still on the waitlist and another 10% had died before receiving a kidney.[18]

KIDNEY TRANSPLANT DONATION

Most transplanted kidneys come from deceased donors. In an attempt to expand this donor pool, new policies allow use of a greater breadth of types of deceased donor kidneys including grafts from older donors and donors after cardiac death.[66] However, a goal to increase the supply of deceased donor kidneys can come into tension with ethically complex questions about the very definition of death and concerns about how to ensure that donation is not in violation of a deceased donor's previous beliefs about bodily integrity.[67] Transplant of a kidney from a living donor leads to better patient and graft survival than transplant from a deceased donor but requires that a healthy person who does not stand to clinically benefit undergo a surgical procedure. There can also be complex psychological and relational dynamics that complicate living donation. A family member or friend may feel pressured or coerced into donation or, alternatively, receipt of a kidney may engender a sense of deep obligation to the donor. Although nondirected living donation can be admirably altruistic, there is concern that donation to a stranger may be motivated by other factors such as hope for media attention, an attempt to contend with depression, or the anticipation of being a part of the recipient's life. There will always be concern about compensation for donation—financial or otherwise—especially as there is no national organization to regulate this practice.[68] Although the United Network for Organ Sharing (UNOS, the contractor currently operating the US Organ Procurement and Transplantation Network) bars directed donation based on race, sex, religion, or national origin, there are many examples of infringements on these directives.[68,69]

To address some of these concerns, a statement from the Live Organ Donor Consensus Group includes requirements for the circumstances of live donation. The donor must be competent, willing to donate, free from coercion, and medically and psychosocially suitable. The donor must also be fully informed of the risks and benefits of this decision, as well as the risks, benefits, and alternative treatment options for the recipient. Further, the benefits to both the donor and recipient must outweigh the risks associated with the donation and transplantation.[70] In 2006, the National Academy of Medicine underlined the need for better empirical evidence about risks to donors to inform this determination.[71] Subsequent work has suggested relatively low, but far from negligible, risk of perioperative complications and long-term kidney health conditions for donors. Donation may also entail less quantifiable burdens such as personal expenses, lost wages, and psychosocial stress. While these risks are concerning in relation to the lack of clinical benefit from a donation procedure, many question the ability of transplant clinicians to adequately gauge nonclinical benefits to donors, who are often the close persons and caregivers for their loved ones with kidney disease.[72,73] Other donors may appreciate the less tangible, but personally important opportunity to demonstrate altruism as part of their effort to live a virtuous life.[74]

KIDNEY TRANSPLANT ALLOCATION

Even as the transplant community pursues opportunities to expand the number of deceased and living kidney donors, efforts are unlikely to entirely close the gap between supply and demand for a growing population of people with kidney failure.[75] There continues to be debate about the most fair and effective approach to distributing deceased donor organs across the population. The UNOS kidney allocation system is guided by the National Organ Transplant Act (NOTA) Final Rule[76] and framed by a set of ethical principles: utility, justice, and respect for persons (Table 80.1).[77] At its most basic, the current kidney allocation system is a waiting line in which candidates for transplant are prioritized by how long they have been on the waitlist. Although this strategy is intended to support fairness, deep underlying disparities in U.S. health care mean that already disadvantaged patients may experience delays in being referred for transplant, being added to the waitlist, and starting to accrue wait time. This shortcoming of a "first-come, first-served" approach likely contributes to disparities in transplants for minoritized racial groups and other underserved populations.[78,79] In an effort to reduce these inequalities, new kidney allocation rules in 2014 included retroactive waiting time credit starting at the time of dialysis initiation.[80] Early reports show improved rates of transplant for Black and Hispanic patients.[80,81] Other revised organ allocation criteria are designed to improve the overall benefit of available organs across the population. For example, a "longevity matching" strategy is intended to pair recipients who are predicted to live the longest after transplant with donor kidneys that are expected to function for many years.[80]

Importantly, while the UNOS organ allocation system continues to be far from perfect, the opportunity to revise and improve allocation is only possible because of the transparency and standardization of the system, as well as rich data collection about patient outcomes. Unfortunately, barriers to transplant earlier in the process of referral, evaluation, and selection for waitlisting remain opaque and more difficult to address. Increasing focus on these upstream processes has prompted calls for data collection about rates of referral across the population.[82] The evaluation and selection process itself also can entail physical, social, and psychological burdens for patients pursuing transplant, especially for older adults and people with multimorbidity who are also less likely to ultimately benefit from the process by receiving a kidney.[83] Clinical practice guidelines informing the evaluation and selection process are limited by an equivocal evidence base.[84,85] The specific approach to selection is deferred to transplant centers, leading to huge variations in how patients are selected.[86] Requirements for a firm base of social support after the transplant procedure are intended to ensure that a patient is able to care for their kidney in the critical months after transplantation but also enact disproportionate barriers to socioeconomically disadvantaged groups.[87,88] A range of other nonclinical barriers to transplant raise further concerns about fairness and bias in the evaluation and selection process.[89,90]

Table 80.1 Ethical Principles in the Allocation of Deceased Donor Kidneys

Principle	Definition and Application	Example From the Kidney Allocation System
Utility	"Allocation should maximize the expected net amount of overall good."[a] Maximize graft survival, years of life added, and/or predicted quality-adjusted life years added.	*Longevity matching*: A score for patient estimated post-transplant survival (EPTS) and deceased donor kidney risk of graft failure (the kidney donor profile index or KDPI) is used to pair patients who are expected to live the longest (top 20% EPTS) with kidneys expected to function for the longest duration (lowest 20% KDPI). This strategy intends to avoid "waste" of kidney years for people who may die with a kidney that would otherwise function for many more years and avoid needing to relist other people who outlive their donor kidney function.
Justice	Allocation should support "fairness in the pattern of distribution of the benefits and burdens of an organ procurement and allocation program … This does not mean treating all patients the same, but it does require giving equal respect and concern to each patient."[a] Allocate donor organs equitably across groups defined by social characteristics (e.g., socioeconomic status, race, ethnicity, gender, geography).	*Retroactive waiting time credit*: Before the new kidney allocation system of 2014, patients began receiving "points" for years they had spent on the waiting list. After 2014, patients accrue time from when they were first added to the waitlist OR from the time that they initiated dialysis, whichever is longer. This change was intended to mitigate disparities in timely referral for transplant evaluation and waitlisting among socially disadvantaged groups.
Respect for persons	"We owe to humans a respect that they should be treated as 'ends in themselves,' not merely as means. This principle embraces the moral requirements of honesty and fidelity to commitments made. Most importantly, respect for persons embraces the concept of respect for autonomy."[a]	The system should respect the rights of those who refuse to donate or receive organs. Allocation rules should be transparent such that individuals can make informed decisions about how and whether they engage in the process.

[a]Organ Procurement and Transplantation Network: OPTN/UNOS Ethics Committee White Paper, Ethical Principles to be considered in the allocation of human organs, <https://optn.transplant.hrsa.gov/resources/ethics/ethical-principles-in-the-allocation-of-human-organs/>; 2018.

PEDIATRIC TRANSPLANT ALLOCATION

Pediatric patients with kidney failure constitute a special population of patients who receive priority for transplant. While this strategy is not without critique, the approach is supported by multiple ethical arguments including a societal consensus around an obligation to protect children, greater need for transplant among children who may fare poorly on dialysis, fairness of allowing all persons similar life opportunities, and utility of selecting patients who are most likely to live for longer periods to receive deceased donor kidneys.[91,92] Nonetheless, scarcity of organs does shape selection criteria for this population. Historically, pediatric patients with intellectual disabilities had been limited in access to transplant because of concern that these patients would be unable to adhere to treatment regiments, may lack adequate social support, and ultimately have lower post-transplant success,[93] though evidence to support these assumptions is lacking.[94] Biases about quality of life for people with intellectual disabilities may further shape transplant clinicians' perception of the benefit of transplant for this population. Disabilities rights advocates point to the unacceptability of a social value criterion in determining transplant allocation, exemplified by the Seattle Artificial Kidney Center's Admission and Policy Committee experience, and emphasize the unique contributions of these patients to their families and society. Evoking legislative precedent—including the Americans with Disabilities Act[95] and Rehabilitation Act[96]—as well as bioethical arguments of justice, the American Academy of Pediatrics and Council on Children with Disabilities have denounced the practice of restricting access to transplant based on intellectual disability as ethically unjustified.[97]

ETHICAL CHALLENGES FOR KIDNEY CARE INSTITUTIONS AND COMPLEX HEALTH SYSTEMS

DISRUPTIVE PATIENT BEHAVIOR AND INVOLUNTARY DISCHARGE FROM A DIALYSIS FACILITY

In recent years, dialysis centers have reported an increasing number of patients with challenging and disruptive behavior. These behaviors may interfere with patients' own treatment, disrupt the smooth running of the center, or affect the welfare of other patients and staff.[98,99] Behavior may include loud outbursts, name calling, or pulling needles in a way that endangers other patients or that threatens immediate severe physical harm to staff or another patient. In the most extreme scenarios, patients risk being discharged from a dialysis center involuntarily.[100]

The medical community endorses strong ethical and legal obligations toward the patient who requires life-sustaining dialysis treatment. At the same time, the ethical principles of respect for patient autonomy, beneficence, nonmaleficence, and justice command an equal moral weight with respect to other patients and staff on the unit. As early as 1990, the "difficult dialysis patient" was identified as one of the top three ethical challenges facing nephrologists.[101] In 1998 when the Renal Physicians Association and the American Society of Nephrology were developing their Shared Decision-Making clinical practice guideline, the Centers for Medicare and Medicaid Services (CMS) advocated for an entire chapter devoted to such patients (Box 80.2).[15]

Over a 4-year period from 2017 to 2020, the Forum of ESRD Networks reported that there were approximately 25,000 dialysis patients for whom involuntary discharges had been intended or implemented. This report was especially troubling because Black Americans comprise roughly one-third of the prevalent national dialysis population but represented 43% of all involuntary discharges and 48% of involuntary discharges among people younger than 45 years of age.[102] For further discussion on health disparities, see Chapter 81.

The first step in approaching a clash between obligations toward a patient with disruptive behavior and need to respect the rights of other patients and staff in the dialysis center is to try to change the situation to reduce conflict that could escalate to an involuntary discharge. The kidney community collaborated in the Decreasing Dialysis Patient–Provider Conflict project to develop resources to help dialysis staff better cope with patient–provider conflict, improve staff–patient relationships, and create safer dialysis facilities.[99] The goal was to increase awareness and improve skills to ameliorate conflict including creating a common language to describe and classify conflicts. Interpersonal conflict resolution is certainly a first step, but underlying causes for involuntary discharges are likely complex and may need to be addressed at the systems level, including cultural gaps between patients and dialysis staff, psychosocial and regional factors, structural racism in kidney care, antiquated practices for treatment of kidney failure, unintended consequences of quality incentive programs, and other perverse incentives. Better understanding of patterns and trends of health disparity in involuntary discharges, based on well-designed surveys and qualitative research, is needed to inform future interventions.[103]

A spectrum of disruptive or difficult behavior ranges from the patient whose behavior harms only the patient themself to one whose behavior endangers other patients and staff in the dialysis unit.[104] Ethical obligation will vary depending on where a patient's behavior falls on this spectrum. Nonadherence alone should not lead to denial of treatment or discharge from the dialysis unit. Even in cases involving verbally or physically abusive patients, discharge from a dialysis unit should be undertaken only as a last resort after the other strategies have been exhausted.[105] This approach allows health care professionals to identify the limited situations in which involuntary patient discharge from a dialysis center is ethically justified.[105]

Box 80.2 Strategies for Dealing with the Disruptive or Difficult Dialysis Patient

Strategies for Working With the Patient

- Learn the patient's story and seek to understand their perspective.
- Identify the patient's goal for treatment.
- Share control and responsibility for treatment with the patient.
  - Educate the patient so that they can make informed decisions.
  - Involve the patient in the treatment as much as possible.
  - Negotiate a behavioral contract.
- Appoint a patient advocate.

Strategies for Preparing the Staff

- Teach the staff not to criticize patients and call them names.
- Have the staff use "reflective listening" to show that the patient has been heard.
- Deal directly with problem behavior; involve the patient, build on the patient's strength, and be clear about who is to do what and when.
- Take a nonjudgmental approach.
- Focus on the issue that started the disagreement.
- Detail the consequences of aberrant behavior in terms that are comprehensible.
- Prepare a behavior contract that specifies what is to be done by the patient and renal team.
- Prepare in advance to manage anger.
- Be patient and persistent.
- Do not tolerate verbal abuse.
- Outline for the staff step-by-step means of coping with agitated and disruptive patients.
- Establish and publicize a grievance procedure.
- After effective resolution of a conflict, follow up with the patient to monitor the patient's progress and demonstrate to the patient the commitment to resolve a conflict.
- Contact law enforcement officials when physical abuse is threatened or occurs.
- As a last resort, consider transferring the patient to another facility or discharge.
- Consult with legal counsel before proceeding with plans for discharge and do not discharge the patient without advance notice and disclosure of future treatment options.
- Contact the end-stage renal disease network if satisfactory resolution has not occurred with the use of these strategies.

From Johnson CC, Moss AH, Clarke SA, Armistead NC. Working with noncompliant and abusive dialysis patients: Practical strategies based on ethics and the law. *Adv Ren Replace Ther.* 1996;3:77–86.

Clinical Relevance

Dialysis centers have been treating an increasing number of patients with disruptive or challenging behavior. Even in cases involving verbal or physical abuse, involuntary discharge from a dialysis center should be undertaken only as a last resort. More research is needed to understand stark racial disparities in involuntary discharge, and interventions are needed to address these health disparities.

FINANCIAL CONFLICTS OF INTEREST IN DIALYSIS CARE

A significant portion of many nephrologists' income is generated from practice in dialysis centers and/or their share in ownership. Ownership offers the opportunity for a nephrologist to direct the mission of a dialysis center to better support patients, but it also risks conflict of interest and potential breach of the physician's fiduciary duty to patients. Decades ago, nephrologist and then *New England Journal of Medicine* editor-in-chief Arnold Relman anticipated this predicament. He warned that the private enterprise system—the emerging medical–industrial complex—had a particularly striking effect on the practice of dialysis, and he urged physicians to proactively separate themselves from any financial participation so as to maintain professional integrity.[106]

As the number of dialysis centers continues to expand and the market becomes saturated, there is intense pressure to recruit patients. A nephrologist who encounters a patient during a hospital admission might encourage the patient to transfer to a different dialysis center in which the nephrologist has a financial interest. This practice of patient solicitation disrespects the physician–patient relationship, informed consent, and continuity of care, and it is a clear conflict of interest. It has been roundly criticized as falling short of ethical conduct.[107,108]

Dr. Relman's strict entreaty to separate practice from financial interests may not be entirely achievable because both nephrologists and dialysis centers do need to maintain healthy financing to continue to provide care. However, medicine as a profession is held to a different ethical standard than other marketplaces. As compared with businessmen engaging in a free-market enterprise, medical providers have a duty to prioritize the well-being of the patient. As formalized by professional societies, nephrologists have a responsibility to give patients evidence-based information about the quality of dialysis centers and treatment options, as well as fully disclose their personal interests if there is a possibility that these could interfere with clinical judgment.[107]

CENTERS FOR MEDICARE AND MEDICAID SERVICES INITIATIVES TO LIMIT FEDERAL SPENDING ON KIDNEY DISEASE

At the time of its implementation in the 1970s, the ESRD program was expected to incur modest costs compared with the value of life-saving dialysis treatment. However, early financial predictions were quickly outpaced by an expanding dialysis population. In 2020, the ESRD program constituted a disproportionate 6.1% of total Medicare expenditures but covered only 1.3% of Medicare beneficiaries.[18] As U.S. health care costs escalate, critics have increasingly raised concern about the moral limits of federal spending on kidney disease and wondered if more good could be done by investing limited resources in other chronic disease programs[109,110] or whether there are opportunities to improve value in the existing program.

Value-based care is one of the CMS's more recent approaches to limit the expense of the dialysis program. In 2019, one of the goals of the Advancing American Kidney Health Initiative was to address the cost of caring for patients with kidney disease and simultaneously to increase the proportion of those receiving treatment aligned with their individual values and preferences, based on informed patient choice. Specifically, this initiative set a goal to reduce the number of Americans developing kidney failure by 25% by 2030 and to increase the number of new patients with kidney failure who were starting on home dialysis or with a kidney transplant. These goals were intended to support upstream kidney health before patients developed disease and to provide this care at a lower cost.[111]

In 2022, Medicare introduced the Kidney Care Choices model as another component of its value-based care initiative for patients with kidney disease. This model incentivizes management of patients by a single set of kidney care providers from the late stages of CKD through dialysis and kidney transplantation. Currently, many patients follow an especially expensive treatment path, often with an unplanned hospitalization to start hemodialysis treatment. A new payment structure incentivizes providers to delay progression of kidney disease, prepare patients for kidney failure including choosing a treatment modality in advance, and encourage referral for transplant. This model is nephrologist led but structured to encourage interdisciplinary integrated care teams and networks designed to manage the entirety of a patient's conditions and also share in the costs or cost savings of this care.[112] Optimally, the model has the potential to be a win-win-win for patients, providers, and the government. Whether and to what extent this model will succeed remains to be seen.

Health care quality has been defined as "the right care for the right patient at the right time." Despite the admirable language of "person-centered treatment options" and "patient choice" in the Advancing American Kidney Health Initiative and the Kidney Care Choices model, these initiatives do not directly address the needs of the approximately 20% of patients with kidney disease who are seriously ill with an impaired quality of life and who have a high risk of death over the course of 1 year.[113] A patient-centered, high-quality approach for seriously ill patients with kidney disease likely necessitates CMS policy changes to realign incentives and 1. exclude patients who choose palliative dialysis from the ESRD Quality Incentive Program, which financially penalizes dialysis centers who fail to meet specified dialysis performance measures; 2. pay for concurrent hospice and dialysis for terminally ill dialysis patients without a requirement for the patient's terminal illness to not be kidney disease-related; and 3. fund demonstration projects to determine optimal ways to implement active medical management without dialysis as a treatment option.

ETHICAL CONUNDRUMS WITH ADVANCING TECHNOLOGY IN KIDNEY CARE

GENETIC ADVANCES IN KIDNEY DISEASE

Better understanding of the genetic factors that predispose to particular diseases has revolutionized detection and counseling for high-penetrance genetic disorders such as autosomal dominant polycystic kidney disease. However, with this information comes complex questions about how to convey risk information to pre-symptomatic carriers, the duty to warn genetically related family members, and how this information should inform reproductive choices. Unfortunately, history demonstrates how mishandling of genetic information can have major adverse effects on communities. In the 1970s, sickle cell screening was implemented before there was a clear understanding of the clinical significance of carrier status, and it caused stigmatization of many patients of African descent, seeding social stigma and distrust of medical providers within this community. In 1997, the World Health Organization proposed international guidelines on many ethical issues in medical genetics. These include protocols for informed consent for testing, disclosure and confidentiality, and presymptomatic testing.[114] The World Health Organization emphasizes the imperative to never use genetic data to stigmatize or discriminate.

Genome-wide sequencing technology has the potential to identify new genetic risk factors for kidney disease on a larger scale. Discovery of the association between APOL1 alleles and risk of progressive kidney disease among people of West African descent was both a critical advance in understanding genetic underpinnings of the disease and also a prime example of the importance of developing an intentional and proactive approach to the complex ethical issues related to genetic advances.[115,116] Discovery of APOL1 helps to explain part of the disproportionately high risk of developing kidney disease among Black Americans, but overlapping genetic risk factors and health consequences of racism are difficult to disentangle. APOL1 has been proposed as a screening tool for transplant donor candidates as a strategy to protect people who are at higher risk for future kidney disease, but this strategy would disproportionately affect Black potential donors and may exacerbate existing disparities in living donation. Ethicists, health services researchers, and scholars studying the myriad health manifestations of racism emphasize the importance of engaging and partnering with representatives of affected communities in determining whether and how genetic information should be used in research and clinical practice.[117,118]

XENOTRANSPLANTATION

For decades, scientific progress in the use of transplantable organs from other animal species has been limited by the hyperacute cellular rejection that results from human-preformed antibodies coming in contact with cellular markers of a foreign species. Leaps forward in genetic modification after the introduction of advanced gene editing methods have opened the possibility of xenotransplantation in the not-too-distant future. These advances have also reinvigorated discussion about ethical issues in xenotransplantation within the medical community and broader public.[119]

Appropriate selection of research participants for early xenotransplantation trials is clinically and ethically complex. For people with heart or liver failure who are not candidates for conventional transplant, participation in a xenotransplant trial may be the only option to preserve life, but for people with kidney failure, who typically have the option to continue dialysis, determination of a benefit-to-risk ratio is less clear.[120] Further, while genetic modification has improved, there remain concerns about transmission of xenozoonotic disease and the possibility that retroviruses could be communicated to the broader population. Research subjects may be limited in their usual rights to exit a research program after transplantation if there was a need to continue to monitor these participants for the development of zoonotic infections that could pose a risk to the broader population. Indeed, experience during the COVID-19 pandemic has underlined the global implications for viral infection control and the World Health Organization has sponsored proposed regulatory requirements for xenotransplantation trials.[121] Finally, while there is clinical precedent in using pig donors for heart valves, tendons, and bones—as well as for a food source—the potential scale and restricted life activities for these animals necessarily to support a porcine organ donation program are of concern for animal welfare activists.

Xenotransplantation may offer a unique opportunity to help the vast number of people with kidney failure who do not currently have access to kidney transplants, but this new technology is not a panacea and prompts new ethical challenges. It is critical to explore these questions in advance of broader adoption and to incorporate voices from the broader community in making decisions that relate to societal values and public welfare.[122]

KIDNEY RESEARCH ETHICS

Dialysis treatment and other kidney therapies have been criticized for insufficient evidence base and lack of innovation for therapies that can be incredibly burdensome for patients. Major advances in the prevention and treatment of kidney failure are grounded in the participation of patients and other volunteers who may not stand to personally clinically benefit from the research. Unfortunately, the history of medical research includes terrible examples of exploitation of research participants, especially populations who were vulnerable or already experiencing discrimination.[5-7] In response, current regulations around protection of human subjects center on minimizing risk to these participants.[8,9] However, evolving research methodologies and technologies present new challenges for research ethics. Large pragmatic trials based in dialysis centers offer the opportunity to examine new treatments and protocols under realistic conditions, but for this scale of work, it may not be feasible to directly consent participants. This approach does limit patient autonomy and may be especially problematic for vulnerable or cognitively impaired groups of patients.[123] Other research projects involve invasive procedures that promise to substantially advance the medical community's understanding of kidney disease but also present real clinical risks to participants and may blur the boundary between research and clinical interventions. For example, one prominent research effort involves performing kidney biopsies for healthy participants and people with kidney disease for whom biopsy would not otherwise be recommended as part of their usual clinical care.[124] There are also persistent disparities in access and participation in research, which may be less inclusive of socioeconomically disadvantaged groups and minoritized populations. These disparities both restrict access to new therapies for these groups and risk perpetuating an evidence base that is not representative of the population. Excitement about new technologies and research methods must be paired with continued vigilance about protection of human research participants, as well as community engagement and involvement to define what this appropriate protection means in evolving research contexts.

A GLOBAL PERSPECTIVE ON ETHICAL ISSUES IN KIDNEY HEALTH CARE

As life expectancy increases worldwide, chronic diseases including diabetes and hypertension become more prevalent. The rate of kidney failure has similarly risen sharply. One systematic review estimates that in 2010, 2.6 million people received dialysis across the globe but nearly the same number died without access to the treatment.[125] The authors further anticipate a doubling of the global prevalence of kidney failure by 2030, with growth largely in regions with limited access to care. Even comparisons across similarly high-income countries show markedly different patterns of dialysis usage. For example, rates of treatment in patients younger than 45 years of age were similar between the United States and Australia, but rates of treatment for patients older than 85 years of age were 41% in the United States compared with 6.8% in Australia.[35] Japan, Taiwan, and Singapore also have patterns of higher rates of dialysis among older adults with kidney failure compared with other high-income countries.[125] This heterogeneity suggests strong political and socioeconomic, health system, and industry forces shaping what is considered appropriate policy and practice in treatment for kidney failure.[126]

Kidney transplant remains a yet more scarce resource with varying availability across countries resulting in a "transplant tourism" culture in which wealthy people may purchase donor kidneys from vulnerable populations in underresourced countries. The Declaration of Istanbul on Organ Trafficking and Transplant Tourism strictly prohibited the sale of organs, as well as travel to a country for the purpose of obtaining an organ, citing violation of the principles of justice and respect for human dignity.[127] With the notable exception of Iran,[128] organ sale is now illegal across the world. Despite this prohibition, black markets flourish. Poor, indebted donors are subjected to dangerous and unregulated surgeries with frequent infectious complications and long-term poor health.[129] They often receive less compensation than advertised, which is rarely sufficient to create lasting changes in their economic situation.

KDIGO guidelines on supportive care draw attention to low-income populations as a distinct group with "choice restriction," emphasizing the importance of developing policy aligned with the realities of differing resources in high- and low-income countries.[130] A World Health Organization analysis of health care interventions for underresourced countries that consider both equitability and cost-effectiveness of potential health care interventions determine dialysis for kidney failure to be a relatively low-priority intervention by these standards.[131] Substantial expense has focused attention on building preventative programs for kidney health in low-resourced countries.[132] PD has also been highlighted as a modality that is well suited to low-resource settings because it requires fewer trained staff members, does not require electricity or large volumes of filtered water, and is importantly potentially cost saving in comparison with hemodialysis (HD).[126,133] Hong Kong's "PD first" initiative, which only offers financial coverage for PD unless there is a contraindication to this treatment, has made Hong Kong one of the only countries to have a greater proportion of patients on PD than HD.[133] As individual nations develop a more connected global community, disparities between high- and low-income countries are more apparent and the question of ethical responsibilities of high-resourced countries becomes more poignant (for further discussion on global inequities, see Chapter 73).

CONCLUSION

The ethical issues facing nephrologists in the 1960s in guiding the development of kidney replacement therapy ushered in the field of bioethics. Decades later, nephrologists, ethicists, and policy makers in the United States and globally continue to be confronted with how to provide an expensive, chronic, life-saving therapy for an increasingly diverse population of people with kidney disease across a wide spectrum of age and health status. Focus has shifted from selecting among patients to receive a scarce dialysis resource to questions around how and when to withhold or withdraw the therapy and support end-of-life care that upholds patient values. Kidney transplantation practices straddle conflicting obligations to improve the life and health of recipients while respecting deceased donors and protecting living donors. The transplant community continues to grapple with how to fairly allocate a scarce supply of deceased donor kidneys across a population of patients. New scientific discoveries and clinical innovations may offer unprecedented treatment options for patients but will require continued investigation into old and new bioethical conundrums. In this context, kidney care providers are likely to continue to be at the forefront of those grappling with ethical challenges in medicine.

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QUESTIONS

1. Which one of the following statements best summarizes the role of shared decision making for patients with chronic kidney disease (CKD)?
   a. Shared decision making is an outmoded concept from the 1980s.
   b. Shared decision making fits well with a disease-oriented approach to CKD patient treatment.
   c. Shared decision making for CKD patients defaults to dialysis modality choices.
   d. Shared decision making is the recognized preferred model for medical decision making.

Answer: d

Rationale: Although shared decision making was introduced in the 1980s, it currently remains the recognized preferred model for medical decision making. It was recommended in 2010 in the second edition of the Renal Physicians Association clinical practice guideline and the American Society of Nephrology Choosing Wisely Campaign in 2012. Shared decision making has been called the “pinnacle of patient-centered” care. It does not fit well with a disease-oriented approach. It concerns much more than dialysis modality.

2. Which one of the following is true regarding informed consent for dialysis for an older patient with stage 5 chronic kidney disease (CKD) with significant comorbid conditions?
   a. An “age-neutral” approach is appropriate to avoid discrimination.
   b. It should include disclosure that dialysis may not confer a survival advantage.
   c. Consent for dialysis is a formality because stage 5 CKD patients want to live.
   d. Physicians are to provide information, not make a recommendation.

Answer: b

Rationale: Multiple studies have found that for older patients with significant comorbidities, dialysis may not confer a survival advantage. Because the research findings particularly address CKD patients older than 75, an “age-neutral” approach has been discouraged. Because of the additional considerations regarding quality of life and survival for older patients, a detailed informed consent discussion is necessary. It is not merely a formality. The ethical model of informed consent requires physicians to make a recommendation, not just provide information.

3. Which one of the following ethical principles justifies patient self-determination regarding deciding to initiate or forgo dialysis?
   a. Beneficence
   b. Justice
   c. Nonmaleficence
   d. Respect for autonomy

Answer: d

Rationale: Respect for patient autonomy is the ethical principle based on the concept that people should be empowered to make decisions for themselves in an uncoerced manner to the extent that they can understand their health condition and treatment options with their attendant risks and benefits. The legal right of patient self-determination is based on this principle. Beneficence, justice, and nonmaleficence refer to ethical principles to promote patient benefit, treat patients fairly, and do no harm, respectively.